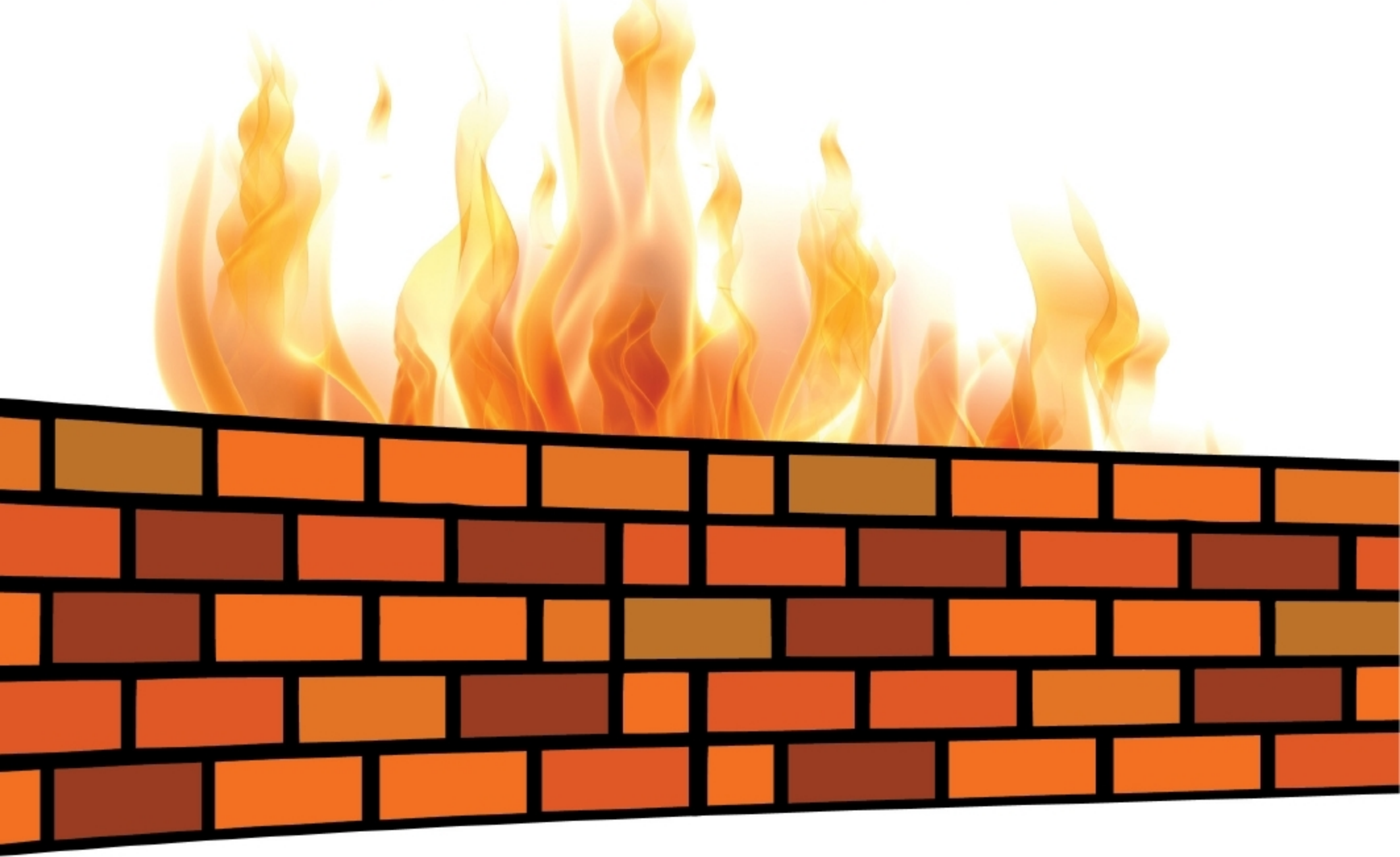


Securing Network Communication *with* firewalld



If you are in need of a dynamically managed firewall with support for network/firewall zones, and with a trust level for network connections or interfaces, then `firewalld` is the tool for you. It is free, open source and built into RHEL 7.

The Linux kernel includes a powerful network filtering sub-system called `netfilter`. This allows kernel modules to inspect every packet traversing the system. This means that any incoming, outgoing or forwarded network packet can be inspected, modified, dropped or rejected in a programmatic way, before reaching components in the user space. That is the main building block for setting up a firewall on a Red Hat Enterprise Linux 7 (RHEL 7) machine.

Interacting with `netfilter`

Although it is theoretically possible for systems administrators to write their own kernel modules to interact with `netfilter`, this is typically not done. Instead, other programs are used to interact with `netfilter`. One of the most common and well-known of these programs is `iptables`. In previous RHEL releases, `iptables` was the main method of interacting with the kernel `netfilter` sub-system.

The `iptables` command is a low-level tool, and it can prove to be inadequate when managing firewalls. In addition, it only adjusts IPv4 firewall issues. Other utilities such as `ip6tables` for IPV6 and `ebtables` for software bridges need to be used for more complete firewall coverage.

Introducing `firewalld`

In RHEL 7, a new method of interacting with `netfilter` has been introduced – it is called `firewalld`, and is a system daemon that can configure and monitor the system's firewall rules. Applications can talk to `firewalld` to request ports to be opened using the DBus messaging system, a feature that can be disabled or locked down. It covers IPv4, IPv6 and, potentially, `ebtables` settings. The `firewalld` daemon is installed from the `firewalld` package. This package is part of a base install, but not part of the minimal install.